

WORKSHEET

Distances on great circles time trial

To calculate the distance between two places on the same meridian of longitude, use the formula

$$\text{Distance} = \frac{\text{change in the angle}}{360} \times 2 \times \pi \times R$$

where R = the radius of the Earth (6400 km)

How many of the following distances can you calculate correctly in 8 minutes?

Express each distance, correct to the nearest km. The first line in the table has been completed as an example.

	Position coordinates of the first place	Position coordinates of the second place	Change in the angle	Distance between the places correct to the nearest km
	10°N, 75°E	50°N, 75°E	40°	$\frac{40}{360} \times 2 \times \pi \times 6400$ = 4468 km (nearest km)
1	20°S, 150°E	30°S, 150°E		
2	5°N, 10°W	30°N, 10°W		
3	6°S, 100°E	24°S, 100°E		
4	28°N, 25°W	2°N, 25°W		
5	48°S, 110°E	5°S, 110°E		
6	12°N, 90°W	0°, 90°W		
7	10°N, 90°W	10°S, 90°W		
8	30°S, 150°W	15°N, 150°W		
9	7°N, 11°E	4°S, 11°E		
10	12°N, 45°E	3°N, 45°E		
11	17°S, 115°E	3°S, 115°E		
12	16°S, 145°E	26°N, 145°E		
13	48°N, 10°W	12°S, 10°W		
14	65°S, 20°W	5°N, 20°W		
15	18°N, 135°W	37°N, 135°W		

Answers

- 1 10° , 1117 km
- 2 25° , 2793 km
- 3 18° , 2011
- 4 26° , 2904 km
- 5 43° , 4803 km
- 6 12° , 1340 km
- 7 20° , 2234 km
- 8 45° , 5027 km
- 9 11° , 1229 km
- 10 9° , 1005 km
- 11 14° , 1564 km
- 12 42° , 4691 km
- 13 60° , 6702 km
- 14 70° , 7819 km
- 15 19° , 2122 km

How did you go?

- 14 or 15 correct
- 12 or 13
- 10 or 11
- Less than 10

Congratulations! You're a champion.

Great work!

You're making progress!

Keep practising. Practice makes perfect.

15.02.

$$1a \ 34-18 \\ = 16^\circ$$

$$b. \text{ Distance} = \frac{16}{360} \times 2 \times \pi \times r$$

$$= 0.04 \times 2 \pi r$$

$$= 0.04 \times 2 \times \pi \times 6400$$

$$= 1787.22 \text{ km.}$$

$$= 1790 \text{ km.}$$

$$2a \ 26-1 \\ = 27^\circ$$

$$b. \frac{27}{360} \times 2 \times \pi \times r$$

$$= \frac{27}{360} \times 2 \times \pi \times 6400$$

$$= 3015.92$$

$$= 3000 \text{ km}$$

$$c. \ S = \frac{d}{t}$$

$$= \frac{3000}{12}$$

$$= 250 \text{ km/hr.}$$

$$3a. 34^{\circ}S - 13^{\circ}N$$

$$= 34 + 13$$

$$= 58^{\circ}$$

because they are opposites

$$3b. \frac{58}{360} \times 2\pi r$$

$$= \frac{58}{360} \times 2 \times \pi \times 6400$$

$$= 6478.662$$

$$= 6480 \text{ km}$$

4.